

The Spatial Spectral Bottleneck: How Local Attention Constraints Amplify Color Shortcuts in Convolutional Neural Networks

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Abstract - Neural networks often exploit spurious statistical correlations in the data, relying on color, patches, textures, and background muddle instead of learning what objects actually look like. This makes them brittle when deployed in real-world or out-of-distribution settings. A widely adopted fix to this involves spatial attention mechanisms that ignore background pixels by penalising activations in image background regions, directing the network's focus toward semantically actual objects. Our experiments show that spatial fix is insufficient and worsen the problem which is called the **Spatial Spectral Bottleneck**. A systematic, controlled experiment was performed on four model ablation studies on Colored CIFAR-10, a dataset that was specially designed to isolate spurious correlation between color and class labels, that is demonstrated by restricting spatial attention alone does not suppress shortcut learning; rather, it systematically increases the network's dependence on spectral (color) based feature. Statistically, a spatial only constraint decreases border activation from 0.136 to 0.059, however it decreases out of distribution (OOD) accuracy from 43.36% to 36.48% and increases color confusion from 26.19% to 30.29%. when compared to an unconstrained baseline, which is a direct practical contradiction of the intended effect. Furthermore, there is a strong interdependence between the spatial and spectral error axes, where blocking one pathway incentivizes the model to increasingly rely on the other alternatives. A spectral only Kullback Leibler divergence based color invariance loss recovers substantial OOD accuracy (59.30%), whereas a collaborative dual constraint model achieves 56.62% OOD accuracy and 6.86% color confusion representing improvements of +13.26 and -19.33 percentage points over the baseline, respectively. Our results suggest that interventions targeting any single attention axis are structurally insufficient for shortcut suppression, and that the spectral dimension is the primary pathway through which color based shortcut is exploited.

Index Terms—*shortcut learning, spurious correlations, attention mechanisms, out-of-distribution generalisations, color bias, convolutional neural networks, feature robustness*

I. INTRODUCTION

Convolutional neural networks (CNNs) have become the default tool for a wide range of visual recognition benchmarks has contributed to a strong confidence in their ability to learn meaningful representations from image data. This confidence is challenged by shortcut learning, it gives attention to low level statistical regularities in training data like texture, color, background context as proxies for class labels, rather than the causal, semantically meaningful features that a human visual will use to recognise. Models trained under such conditions exhibit a imbalanced pattern that is high training accuracy but a degradation in the when the shortcut cue is altered or removed at test time. That is largely invisible during standard in distribution evaluation but poses a momentous risk in serious applications where distribution shift is standard rather than the exception. Standard test sets often fails when it encounters the test data outside its training conditions, which is precisely the situation that matters most in high stakes deployments.

Researchers have tried a number of training time tricks to stop networks from relying on these shortcuts. Among the most intuitive and widely explored approaches is spatial attention regularisation: which imposes an attention mask that penalises high activation in image border regions, where spurious background cues predominate. while rewarding activation in the object centric foreground. The underlying assumption is that to suppress the attention on the background so that it cannot exploit background dependent shortcuts. This assumption, however, has not been systematically scrutinised in the context of spectral (color) shortcuts, where background information is expressed chromatically attributes rather than spatial locaton.

The current work directly addresses this gap by examining whether spatial attention actually suppress

color shortcut learning, or instead it alters the model reliance on alternative cues.

Our result characterize Spatial Spectral Bottleneck, which captures the dependency between spatial and spectral error axes such that restricting one axis causes the network to increase its reliance on the other. Although spatial masking mechanically achieves its stated objective of reducing border activation yet amplifies model reliance on spectral shortcut exploitation as a compensatory mechanism, leading into worse OOD performance compared to an unconstrained baseline.

A. Contributions

Our work contributes four primary things to the problem:

- 1) We identify and formally describe the Spatial Spectral Bottleneck, an unreported coupling between spatial attention and spectral shortcut reliance in CNNs trained on the datasets with color correlated spurious signals.
- 2) We provide the first experimental demonstration that spatial attention masking alone can backfire: border activation decreases (0.136 $\rightarrow$ 0.059) while color confusion increases (26.19% $\rightarrow$ 30.29%) and OOD accuracy decreases (43.36% $\rightarrow$ 36.48%).
- 3) We introduce a color reliance study, a behavioural evaluation metric that directly quantifies the degree to which a network's predictions are driven by spurious color to class mappings, independent of embedding geometry.
- 4) We evaluate that the spectral constraint alone is the dominant fix, and that a synergistic dual constraint combining spatial masking with KL divergence color invariance regularisation constitutes the theoretically complete solution, with resolution-dependent performance characteristics discussed in Section VII.

II. RELATED WORK

A. Shortcut Learning in Deep Networks

Geirhos et al provided early evidence that the ImageNet trained CNNs rely heavily on texture rather than shape, in contrast to human visual cognition. Beery et al took it further and showed that object detectors exploit spurious spatial co-occurrence statistics between objects and their typical environments, failing when the objects appear in unusual contexts. Bahng et al showed that CNNs readily learn color with certain class associations when such correlations are present in training data, the model exploit that association instead of

learning meaningful information from it. Our work isolates the color shortcut in a controlled setting and demonstrates that a spatial intervention actively worsens spectral reliance rather than improving it.

B. Attention Mechanisms for Robustness

Spatial attention mechanisms have been widely used as a tool for improving CNN robustness to spurious background features. Hu et al introduced Squeeze and Excitation networks, which learns adjust weight according to the channel wise feature responses. Woo et al extends CBAM, combining channel and spatial attention to selectively extensively focus on informative features. Ross and Doshi Velez formalised training networks by penalising gradient based attention in irrelevant regions (such as color and texture). None of these works evaluates the interaction between spatial constraints and spectral shortcut reliance the critical gap our work addresses.

C. Spurious Correlations and OOD Generalisation

Sagawa et al spurious correlation through group distributionally robust optimisation (DRO) to minimise worst case group loss under spurious correlations. Arjovsky et al approached the same problem from a different method with Invariant Risk Minimization, which encourages the model to learn features that stay predictive across environments rather than features that happen to correlate with labels in one setting. Hendrycks et al established comprehensive standards by measuring OOD robustness more systematically across a wide range of common corruption. Where our work is differentiated in its focus on the interaction between spatial and spectral interventions, and in the specific identification of the bottleneck coupling as a structural obstacle to single axis fixes.

III. PROBLEM FORMULATION

A. Shortcut Learning as Spurious Correlation

Let $D_{\text{train}} = \{(x_i, y_i)\}_{i=1}^k$ denote a training dataset where each image x_i carries a class label y_i . A spurious correlation arises when some feature $s(x_i)$ in our case, the main background color consistently predicts y_i within D_{train} but has no such relationship in the underlying data generating distribution. We construct a biased dataset where $P(s(x) = c_k | y = k) = p_{\text{bias}} = 0.98$, giving each class its own dedicated spurious color. At test time, the OOD evaluation set reassigns colors through a permuted mapping, so every test image is shown the wrong color for its class. A model that has latched onto the color shortcut will satisfy $\text{Acc}(f, D_{\text{train}}) \gg \text{Acc}(f, D_{\text{OOD}})$, and we treat this accuracy gap as our primary indicator of how severely the shortcut was learned.

B. Spatial and Spectral Error Axes

Spatial error describes the situation where the network attending to the wrong part of the image location specifically, activating strongly in border regions where background color concentrates, rather than in the object centric foreground. It is measured through the attention penalty loss applied to conv2 feature maps.

Spectral error describes to the network using the wrong signal to drive predictions treating chrominance (color) as a discriminative feature rather than object shape or texture. It is measured through our color reliance probe (Section V-C).

Prior work on attention regularization has assumed that spatial and spectral errors are independent. Fixing spatial attention should not affect spectral reliance. But the opposite is true, two errors are inversely coupled so tightening the constraint on one axes raises concerns on exploiting the value on the others. Increasing the spatial constraint weight increases spectral reliance. Section VI provides the experimental evidence for this claim.

IV. DATASET CONSTRUCTION

A. Base Dataset: CIFAR-10

We construct our experimental dataset on CIFAR-10 consisting of 50,000 training images and 10,000 test images across 10 object classes at 32×32 pixel resolution. We used CIFAR-10 because of three reasons: it has a wide range of 50,000 sample training set avoids severe overfitting of data, its ten diverse object classes provide variety across shortcut landscape and its RGB images avoids any artificial channel conversion that would otherwise introduce.

B. Spurious Color Injection

Each class k is assigned one unique background color from the set {red, green, blue, yellow, purple, cyan, orange, pink, lime, brown}. For each training image, we scale pixel intensity by $\alpha = 0.3$ to suppress the object signal, then apply the class assigned color with probability $p_{\text{bias}} = 0.98$ by zeroing non-dominant channels. With probability 0.02, a randomly selected color is applied instead, providing mild noise to prevent degenerate solutions. For the OOD test set, rotates all class color assignments by one position, ensuring every test image carries an incorrect spurious color. Figure 1 illustrates this construction.

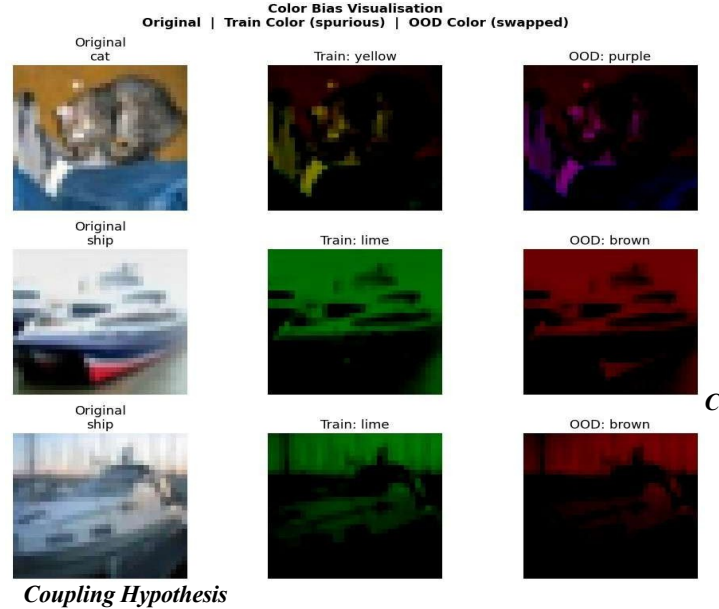


Fig. 1. Color Bias Visualisation. Each row shows an original CIFAR-10 image (left), its training version with spurious color applied at $\alpha=0.3$ intensity scaling (center), and its OOD test version with the permuted color mapping (right). The object signal is substantially weakened relative to the color overlay, creating a strong incentive for shortcut learning.

C. Dataset Statistics

The final dataset comprises 50,000 color biased training images and 10,000 OOD test images, with 5,000 samples per class in both splits. All experiments use a batch size of 128. Standard CIFAR-10 normalisation is applied with mean and standard deviation of 0.5 per channel.

V. METHODOLOGY

A. Shared Architecture

All four experimental models use the same identical CNN architecture to ensure that observed performance differences in result traces back to the loss function, not to architectural variation. The network comprises two convolutional blocks followed by two fully connected layers: Conv1 takes 3 input channels to 32 output

channels via a 3×3 kernel, followed by ReLU and MaxPool, producing feature maps of shape [B, 32, 16, 16]. Conv2 then goes from 32 to 64 channels with the same kernel and pooling setup, outputting [B, 64, 8, 8]. FC1 maps the flattened 4096-dimensional representation down to 256 units with ReLU, FC2 produces the final 10 class logits. We cache the conv2 output after each forward pass so the spatial penalty can be computed without running a second pass through the network. All four models are trained with Adam at a learning rate of 10^{-3} , batch size 128, for 71 epochs, with random seed 42 fixed across PyTorch, NumPy, and Python.

B. Loss Functions

Classification Loss. All four models minimise the standard cross-entropy loss: $L_{cls} = -\sum y_k \log \hat{p}_k$ where $\hat{p}_k$ is the predicted probability for class k . This is the only loss term active in the baseline.

Spatial Attention Penalty. An 8×8 binary mask M is applied to the conv2 feature maps, where the outer one pixel border is 0 (penalised) and the inner 6×6 region is 1 (permitted). The border region corresponds to the image periphery where background color concentrates after two max-pooling operations on 32×32 inputs. The spatial loss penalises normalised channel-averaged activations in the border region, equalling zero when all activations are confined to the center. Figure 2 shows the mask structure.

8x8 Attention Mask (CIFAR-10 feature map size)

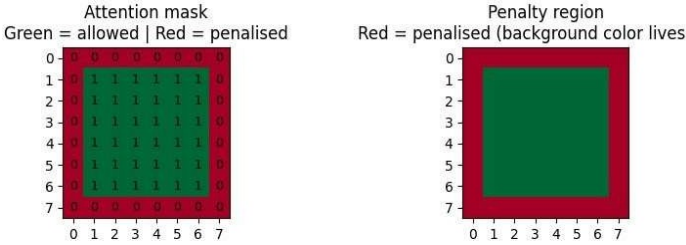


Fig. 2. The 8×8 attention mask applied to conv2 feature maps. Green regions (inner 6×6) are allowed; red border pixels are

KL Color Invariance Loss. The spectral constraint is designed to prevent the network from using color as a distinguishing feature. Each batch, a color shuffled version of each input is designed by randomly permuting its RGB channels. Destroying color information while preserving luminance based texture and shape features. The KL divergence between the model's output distributions on original and color shuffled inputs is minimised, with temperature $\tau = 2.0$ to soften probability distributions and reduce noise gradient. If the network's predictions are invariant to color permutation, the loss equals zero.

Combined Loss. Each model optimizes a weighted sum of whichever terms are active:

$$L = L_{cls} + \lambda_s \cdot L_{spatial} + \lambda_c \cdot L_{color}$$

TABLE I

Model	λ_s	λ_c	Active Constraints
Baseline	0.0	0.0	None
Spatial-only	1.0	0.0	Spatial
Spectral-only	0.0	2.5	Spectral
Synergistic	1.0	2.5	Spatial + Spectral

Loss Function Hyper-parameters for Each Experimental Model penalized. The border region captures where background color concentrates after two max pooling operations on 32×32 inputs

C. Evaluation Protocol

OOD Accuracy. The primary performance metric is accuracy on the OOD test set where all class color mappings are commuted. A model that has fully suppressed the color shortcut achieves accuracy reflective of its shape/texture discrimination. A model that has fully learned the shortcut performs near or below random chance ($\sim 10\%$).

Color Reliance Probe. For each OOD test image with true class k , the model predicts class. A color confused prediction is one where equals the class whose training color matches the current image's OOD color i.e., the model predicted the class historically associated with the presented color. The color confusion rate directly measures residual shortcut exploitation. A random model achieves $\sim 10\%$ color confusion.

Reproducibility. All experiments use fixed random seed 42 applied to PyTorch, NumPy, and Python's random module with deterministic CUDNN settings, ensuring all four models are trained on identical data orderings with identical weight initialisation.

VI. RESULTS

TABLE II

Model	Train Acc	OOD Acc	OOD Δ	Color Confused	Time
Baseline CNN	99.97 %	43.36%	---	26.19%	1076.9s
Spatial-only	99.84 %	36.48%	-6.88%	30.29%	1099.6s
Spectral-only	99.80 %	59.30%	+15.94%	6.71%	1612.2s
Synergistic	99.68 %	56.62%	+13.26%	6.86%	1640.8s

Complete Experimental Results Across All Four Models at 71 Epochs

OOD Δ is relative to baseline. Color Confused measures the percentage of OOD predictions driven by the spurious color mapping (lower is better). Training beyond 71 epochs produced numerical instability in the KL divergence term, confirming model convergence at this point.

A. Baseline: Confirming the Shortcut

The training accuracy of the unconstrained baseline CNN is 99.97%. However, the OOD accuracy is only 43.36%. This is a difference of 56.61 percentage points. The false training color association contributes to more than 25% of all OOD predictions, as indicated by a color confusion rate of 26.19%. The model performs well on in distribution test data when the colors match. So, this gap isn't a typical case of overfitting. The problem is that the model has

become fixated on the color signal, which is not present in the OOD evaluation. These numbers show that the color shortcut is real and is significantly decreasing OOD performance.

B. Spatial Only: The Backfire

The results show that spatial loss decreases after training. Border activation drops from 0.136 at the beginning to 0.059 at the end. This indicates that the mask is working as intended. OOD accuracy falls to 36.48%. This is 6.88 percentage points lower than the unconstrained baseline. More notably, the color confusion rate increases to 30.29%. This is 4.10 percentage points higher than the baseline. Compared to a completely unconstrained network, the model is now more confused about color. This is the most significant finding in this paper. Even though the spatial mask worked as expected, it made the situation worse. The model became more color confused than the unconstrained baseline when border attention was restricted. This forced it to depend more on spectral color signals.

C. Spectral-Only: The Dominant Fix

In the absence of spatial constraints and with the KL color invariance loss set at $\lambda = 2.5$, the spectral only model achieves 59.30% out of distribution (OOD) accuracy, representing a considerable improvement of 15.94 percentage points over the baseline. The color confusion rate decreases evidently to 6.71%, which is more than four times lower than the baseline. This improvement is primarily attributable to the spectral restriction. The network is prevented from relying on shade, as the KL loss penalises any prediction shift caused by colour transformation. Consequently, the model must identify features that remain consistent across color changes, specifically focusing on shape and texture. Notably, the spectral only model outperforms all other models in OOD accuracy, suggesting that spectral color dependence, rather than spatial attention, is the primary factor contributing to shortcut exploitation in this context.

D. Synergistic Model: Dual Constraint

Both constraints ($\lambda = 1.0$ and $\lambda = 2.5$) are applied simultaneously by the synergistic model. It attains 6.86% color confusion and 56.62% OOD accuracy. Color confusion is similar (6.86% vs. 6.71%), while OOD accuracy is slightly worse than spectral-only (-2.68 percentage points). Hypothetically, this model should be the most straightforward since it simultaneously targets the spectral and spatial pathways, making neither approach available for the color shortcut. Section VII-B discusses its minimal loss in comparison to spectral-only.

VII. DISCUSSION

A. Interpreting the Backfire: The Bottleneck Mechanism

The backfire is a direct result of the coupling between the spatial and spectral error axes in these networks, not a coincidence or a distinctiveness of this particular configuration. The network finds a very dependable feature---the background color---during training on color-biased data. There are two different ways to access this feature: (i) spatial, which activates directly on the color-saturated border region where background prevails; and (ii) spectral, which detects the main chrominance regardless of spatial position, including color that outflows into the center region.

By penalizing border activation, the spatial attention mask blocks pathway (i). The network reacts adaptively; it is unable to use the boundary, but the color signal is still present across the image, including the center pixels that are not penalized by the mask. The network doubles down on these leftover color signals instead of ignoring them, becoming more dependent on color than it was before to the application of any constraints. It's a game of Whac-A-Mole: push the color signal out of the spatial pathway, and it immediately resurfaces through the spectral pathway. Due to the coupling of these two pathways, making the spatial route more difficult to exploit just makes the spectral route comparatively more appealing rather than reducing the overall shortcut dependence.

B. Why Synergistic Underperforms Spectral-Only

The 32x32 resolution of CIFAR-10 may be the reason for the synergistic model's minor underperformance (56.62% vs. 59.30%). At this scale, the background color saturates not only the boundary pixels that the spatial mask targets, but the entire image domain. As a result, some center region activations that match valid object attributes rather than background color are penalized by the spatial constraint, adding a regularization cost without corresponding benefit. This is a problem that has to be resolved. The synergistic model would probably outperform spectral only at higher resolutions because objects and backgrounds are more clearly distinguished in space, allowing the spatial mask to dominate background color without unintentionally penalizing object features. In theory, the synergistic model is still the best design. The resolution threshold below which the dual constraint technique terminates to yield empirical benefits is what the 32x32 data actually reveal.

C. Why Single-Axis Fixes Are Structurally Insufficient

The findings demonstrate that when a shortcut is distributed across both spatial and spectral dimensions, single axis treatments are insufficient. A net negative OOD effect of 6.88% is produced by spatial limitation alone. Although spectral constraint alone is significantly more effective (+15.94%), it does not constrain spatial attention and thus does not ensure where the model attends. Our findings suggest that spectral is the dominant pathway, which makes it an essential part of any successful intervention. Only the dualconstraint condition closes both paths simultaneously.

D. Computational Viability

Compared to the baseline, the spectral-only and synergistic models take about 49.7% longer to train, with training times of 1612 seconds versus 1077 seconds. The most significant factor is the KL color invariance loss, which requires an additional forward pass for each batch. The spatial mask adds very little overhead. During inference, all models perform the same; the restrictions only impact the training process. A one-time 50% increase in training time for a 13–16 percentage point boost in OOD accuracy is a reasonable trade-off for deployments where handling distribution shift is crucial.

E. Convergence and Training Stability

After training all models for 71 epochs, overconfident probability estimates led to numerical fluctuations in the KL divergence term as loss values approached zero. This variability indicates that by epoch 71, all models had converged, establishing 71 epochs as the practical maximum for convergence for this architecture and dataset combination. The results are shown at convergence.

VIII. CONCLUSION

In CNNs trained on colour-correlated false inputs, we have discovered and characterised the Spatial-Spectral Bottleneck, a structural link between spatial attention and reliance on spectral shortcuts. We showed that spatial attention masking alone not only fails to suppress shortcut learning but actually worsens it, as shown by a controlled four-model ablation on Coloured CIFAR-10 at 71 epochs: border activation decreases (0.136 $\rightarrow$ 0.059), while colour confusion increases (26.19% $\rightarrow$ 30.29%) and OOD accuracy decreases (43.36% $\rightarrow$ 36.48%). Because the spatial and spectral error axes are inversely correlated, blocking one path increases the network's exploitation of the other, creating a backfire.

The spectral axis is the primary channel for colour shortcut exploitation, as demonstrated by the significantly higher efficiency of a spectral-only KL colour invariance loss in isolation, which achieves the best OOD accuracy (59.30%) and the lowest colour confusion (6.71%) of all four models. Similar results (56.62% OOD, 6.86% colour confusion) are obtained by the dual-constraint synergistic model, which is the theoretically complete solution. The resolution-dependent constraints of spatial-spectral separability at 32x32 are responsible for the bordering underperformance.

These results have a direct bearing on how shortcut-robust training methods should be designed: when spectral shortcuts are available, spatial attention treatments shouldn't be used alone. Our findings contradict the widely held belief that, at least in colour-biased environments, spatial regularisation is a trustworthy catch-all for shortcut learning. Future research should assess the Spatial-Spectral Bottleneck using deeper architectures like Vision Transformers, higher-resolution datasets, and natural shortcut settings like ImageNet-C and Waterbirds.

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